Ratios and matching amounts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

6. 300 ÷ 100

7. 72 ÷ 6

8. Yellow : purple is 2 : 5. There are 28 beads. Find both counts.

9. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

10. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

11. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

12. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

13. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

14. Red: blue is 4 : 4. Complete (4, 4), (16,__).

15. A mix has 10 raisins and 8 nuts. Divide both counts by 2; write the new ratio.

16. A recipe uses 6 cups oats for 14 cups seeds. How many cups seeds go with 9 cups oats?

17. Gold: silver beads are 4 : 3. There are 21 beads. Find each count.

18. Orange: purple tiles are 3 : 8. What fraction of all tiles is orange?

CHALLENGE / Explain your plan and check the result

C1. A table lists (bags, marbles): (2, 14), (5, 35), (8, 54). One pair breaks the ratio. Identify it, repair it, and give the pair for zero bags.

C2. Team A has red: blue ribbons 2 : 3. Team B has 3 : 5. Each team has 40 ribbons. Find red ribbons for each team and compare the red fractions.